

Introduction (graduate level)

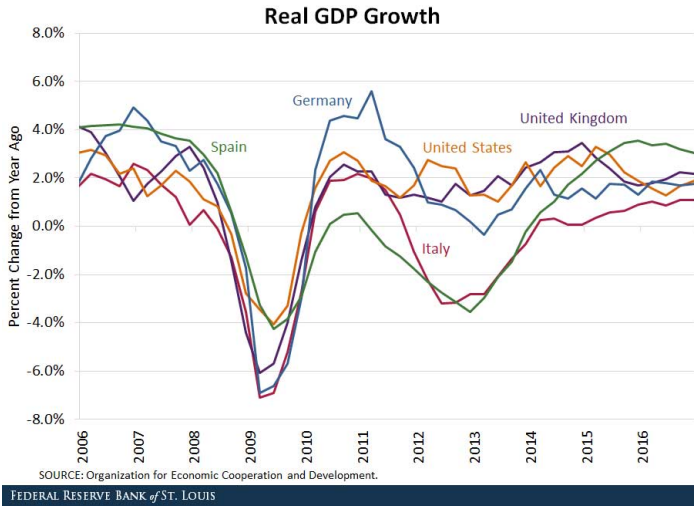
Patterns and determinants of trade: gravity equations

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Introduction

- The economic development of countries is interconnected.



- One channel for this connection is trade.

Why study trade?

Trade is a centuries-long topic in economics.

- The first formal economic model was about trade, written by the Scottish philosopher David Hume in “Of the Balance of Trade.”
- In 19th century Britain, there was a heated debate about whether or not to promote free trade (as opposed to the Corn Laws).
- David Ricardo formalised the idea of comparative advantage in support of free trade.
- The Corn Laws were repealed in 1846 after Ricardo's death. His free trade paradigm became public policy.

Why study trade?

- Trade has been a fundamental aspect of human interaction for thousands of years.



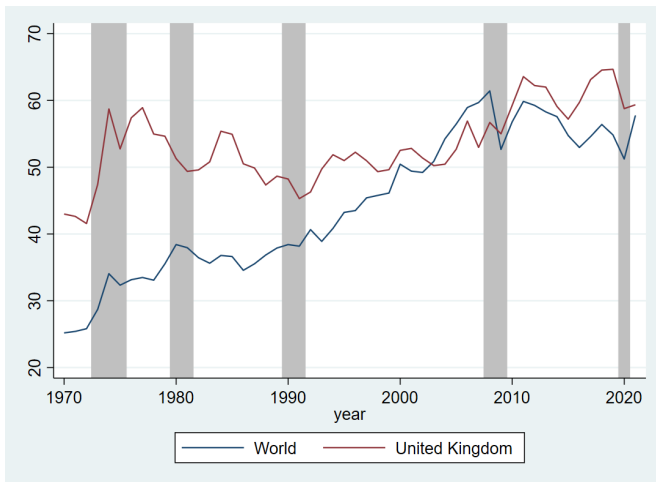
Cuneiform tablet: caravan account ca. 20th–19th century B.C. Kültepe.

Source: Spar (1988)

- long distance travel by Assyrian merchants between Mesopotamia and Anatolia (6-8 weeks)
- trade a large number of tin and textiles for gold and silver
- Mesopotamian tools used to record transactions: cuneiform writings in clay tablets
- The text recorded investment in trade items (tin and textiles), expenditures such as clothing and wages for guides.

Why study trade?

- Trade has become increasingly important.



Trade as percentage of GDP (shaded area indicate UK recessions)

Data source: World Bank national accounts data, and OECD National Accounts data

Why study trade?

- Trade $\implies$ Economic development
 - There are gains from trade.
 - Reason 1: comparative advantage (Lecture 2)
 - Reason 2: inequality in endowment
 - Reason 3: specialisation and economy of scale (Lecture 5)
- Trade $\implies$ Inequality
 - Within a country, there are winners and losers from trade.
 - Free trade hurts owners of immobile factors of production (Lecture 3).
- How much and what do countries trade in practice? (from classical models to new trade theories)
 - Balance between barriers to trade and drivers of trade (Lecture 4)
 - Firm heterogeneity (Lecture 6)

Our plan for today

- Today we start by looking at patterns of trade.
- We will introduce the gravity equations (theories and estimations):
 - the naïve gravity equations,
 - the general gravity.
- For empirical papers that incorporate insights from gravity, we are going to look at:
 - Pascali (2017),
 - Donaldson and Hornbeck (2016),
 - Martincus and Blyde (2013).

Patterns of trade: Gravity equations

- **Gravity equations** can be used to characterise trade patterns and analyse the determinants of trade.
- They were first introduced by Jan Tinbergen (the first Nobel Prize winner in Economics) in 1962.
 - It was not theory-based at the time but is now micro-founded by trade models.
- The equations relate bilateral trade flows between any two countries to their economic sizes and trade barriers (e.g. the distance) between them.
- The name of the equations comes from the analogy to Newton's Law of Universal Gravitation. Paul Krugman refers to them as an example of “social physics”.

“Naïve” gravity equations

“Naïve” gravity equations:

$$X_{ni} = GY_i^a Y_n^b \phi_{ni} \quad (1)$$

- X_{ni} is export from country i to country n .
- G is the “gravitational constant”.
- Y_i and Y_n are GDPs of country i and country n respectively.
- ϕ_{ni} is the bilateral accessibility term,
 - which combines trade costs (e.g. proxied by distance) with the respective elasticity.

“Naïve” gravity equations

If $a = b = 1$ and we use the inverse of distance to proxy for accessibility, the gravity equation becomes:

$$X_{ni} = G \cdot \frac{Y_i \times Y_n}{Dist_{ni}}. \quad (2)$$

It indicates that

- Trade between any two countries increases with **and is proportional to** their GDPs.
- Bilateral trade diminishes with **and is inversely proportional to** distance.

Does this match the data?

Does gravity match the data?

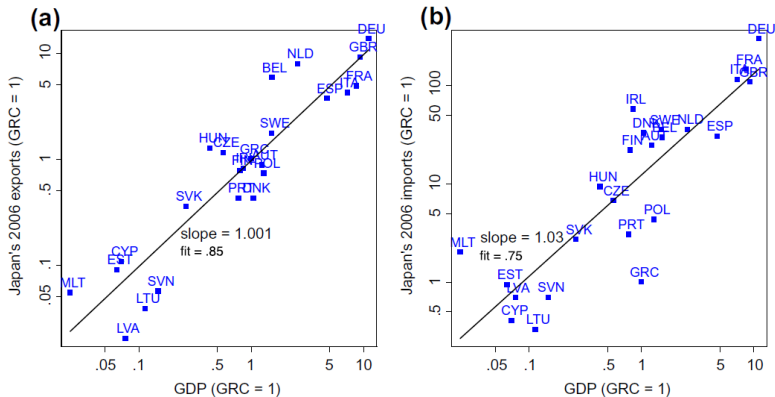


Figure 3: Trade is proportional to size

Source: Head and Mayer (2014)

Estimate gravity equations with data

“Naïve” gravity equations:

$$X_{ni} = GY_i^a Y_n^b \phi_{ni}$$

By taking natural logs on both sides (and use the inverse of distance to the power of θ to proxy for ϕ_{ni}), we have:

$$\ln X_{ni} = g + a \ln Y_i + b \ln Y_n - \theta \ln D_{ni}. \quad (3)$$

- where $g = \ln G$,
- D_{ni} is the distance between country i and n ,
- log-log coefficients are elasticities.
 - θ is the distance elasticity.

Does gravity match the data?

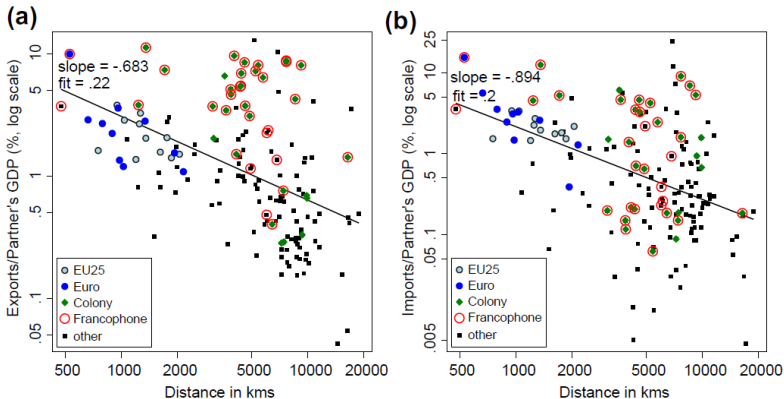


Figure 4: Trade is inversely proportional to distance

Source: Head and Mayer (2014)

Note: We can rewrite the gravity equation as $\frac{X_{ni}}{Y_i} = GY_n \phi_{ni}$ if $a = 1$ and $b = 1$. Then estimate $\ln \frac{X_{ni}}{Y_i} = c - \theta \ln D_{ni}$.

The gravity equation: Recent development

- Despite the success of the “naïve” gravity equations in matching data, we lack understanding of their theoretical underpinnings.
- Recent developments in trade models laid the theoretical foundations for gravity equations.
- We can write more general expressions of gravity equations that are compatible with a host of trade models:
 - that make different assumptions about market structure, production technologies and firm heterogeneity.
- Fixed effects estimations replace estimations using GDPs.

General gravity

General gravity comprises the set of models yielding bilateral trade equations as:

$$X_{ni} = GS_iM_n\phi_{ni}. \quad (4)$$

where

- S_i captures exporter/origin effect,
- M_n captures importer/destination effect,
- S_i and M_n can be functions of observables,
- $0 \leq \phi_{ni} \leq 1$ is the bilateral accessibility term,
- G is the gravitational constant (in the cross-section).

Assumptions:

- Each term enters multiplicatively.
- Third-country effects, if any, are mediated via S_i and M_n .

Structural gravity

A subset of general gravity has the following form and is referred to as the structural gravity:

$$X_{ni} = \underbrace{\frac{Y_i}{\Omega_i}}_{S_i} \underbrace{\frac{X_n}{\Phi_n}}_{M_n} \phi_{ni}. \quad (5)$$

where

- $Y_i = \sum_n X_{ni}$ is the value of exporter's production (to all destination countries),
- $X_n = \sum_i X_{ni}$ is the value of importer's expenditure (on all source countries),
- $0 \leq \phi_{ni} \leq 1$ is the bilateral accessibility term.

Ω_i and Φ_n are “multilateral resistance” terms defined as:

$$\Omega_i = \sum_l \frac{\phi_{li} X_l}{\Phi_l}; \quad \Phi_n = \sum_l \frac{\phi_{nl} Y_l}{\Omega_l} \quad (6)$$

A large set of trade models can derive structure gravity as Equation 5.

Producer market access

How to derive Equation 6?

- First, the sum of country i 's exports to all destinations equals its production Y_i :

$$Y_i = \sum_n X_{ni} = S_i \sum_n \frac{\phi_{ni} X_n}{\Phi_n}. \quad (7)$$

Solving for S_i , we have:

$$S_i = \frac{Y_i}{\Omega_i}, \quad \text{where} \quad \Omega_i = \sum_l \frac{\phi_{li} X_l}{\Phi_l}. \quad (8)$$

- Ω_i is also known as the producer market access.
 - The sum of the size of all destination markets (given by their expenditures, X_l), weighted by the bilateral accessibility term ϕ_{li} and the competitiveness of the destination market measured by Φ_l .

How to derive Φ_l ?

Consumer market access

- Let us define π_{ni} as country n 's expenditure share on goods from country i , in other words, we have: $X_{ni} = \pi_{ni}X_n$. Using Equation 5, we can write:

$$\pi_{ni} = \frac{S_i \phi_{ni}}{\Phi_n} \quad (9)$$

- The second accounting identity is that the expenditure shares add up to one. We have:

$$\sum_l \pi_{nl} = \sum_l \frac{S_l \phi_{nl}}{\Phi_n} = 1 \rightarrow \Phi_n = \sum_l S_l \phi_{nl}. \quad (10)$$

Using Equation 8, we have:

$$\Phi_n = \sum_l \frac{\phi_{nl} Y_l}{\Omega_l}, \quad (11)$$

which is the consumer market access:

- the sum of the production of all source countries, weighted by bilateral accessibility and producer market access.
- It measures the degree of competition in country n .

Producer and consumer market access

If we assume balance trade (i.e. $X_i = Y_i$) and symmetric trade costs ($\phi_{ni} = \phi_{in}$), we have

$$\Omega_i = \Phi_i \quad \text{and} \quad S_i = M_i; \quad (12)$$

We have a symmetric gravity equation.

Example of structural gravity

The Armington Model:

- A world with finite number of countries, with each producing a unique good (the Armington assumption).
- Iceberg trade cost τ_{ni} :
 - for one unit of a good to arrive in destination i , origin n must ship τ_{ni} units.
 - As a result, $p_{ni} = p_i \tau_{ni}$.
- Constant elasticity ($\sigma > 1$) of substitution (CES) utility function:

$$U_n = \left(\sum_i (A_i q_{ni})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \quad (13)$$

where A_i is a utility shifter and q_{ni} is consumption of country n of the product from i .

Armington model

The utility maximisation problem:

$$\max_{q_{ni}} \left(\sum_i (A_i q_{ni})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad \text{s.t.} \quad \sum_i q_{ni} p_{ni} \leq Y_n \quad (14)$$

Solving for the maximisation problem, we can obtain:

$$q_{ni} = A_i^{\sigma-1} p_{ni}^{-\sigma} Y_n P_n^{\sigma-1} \quad (15)$$

where $P_n^{\sigma-1}$ is the Dixit-Stiglitz price index:

$P_n^{\sigma-1} = \sum_l \left(A_l^{\sigma-1} p_{nl}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$. Multiplying Equation 15 by the price p_{ni} and using iceberg trade cost, we have:

$$X_{ni} = A_i^{\sigma-1} p_i^{1-\sigma} \tau_{ni}^{1-\sigma} Y_n P_n^{\sigma-1} \quad (16)$$

Therefore, $S_i = A_i^{\sigma-1} p_i^{1-\sigma}$, $\phi_{ni} = \tau_{ni}^{1-\sigma}$, $M_n = Y_n P_n^{\sigma-1}$.

- This is an example of structural gravity using demand-side derivations. Lecture 4 will show a supply-side derivation.

Estimation of gravity

Equation 5:

$$X_{ni} = \underbrace{\frac{Y_i}{\Omega_i}}_{S_i} \underbrace{\frac{X_n}{\Phi_i}}_{M_n} \phi_{ni}.$$

Equation 6:

$$\Omega_i = \sum_l \frac{\phi_{li} X_l}{\Phi_l}; \quad \Phi_n = \sum_l \frac{\phi_{nl} Y_l}{\Omega_l}$$

Method 1: Iterative structural estimation

1. Set the initial values: $\Omega_i(0) = 1$, $\Phi_n(0) = 1$.
2. Obtain the values of vector ϕ_{ni} using Equation 5.
3. Use ϕ_{ni} to update the vector of values of Ω_i and Φ_n using Equation 6: $\Omega_i(1)$, $\Phi_n(1)$.
4. Go back to (2) to obtain a new ϕ_{ni} .
5. Repeat the process until the values converge (i.e., $\Phi_n(t+1) = \Phi_n(t)$ and $\Omega_i(t+1) = \Omega_i(t)$).

Estimation of gravity: Fixed effects

Method 2: Fixed effects estimation

- We can take natural logs on Equation 4 and obtain:

$$\ln X_{ni} = \ln G + \ln S_i + \ln M_n + \ln \phi_{ni}. \quad (17)$$

- The naïve gravity equations use GDPs, Y_i and Y_n , for S_i and M_n .
- We can instead use fixed effects for S_i and M_n (common practice now).
- With time series data, use time-varying importer and exporter fixed effects.
- With multiple industries data, use industry-specific importer and exporter fixed effects.
- Additional advantage of fixed effects estimation: control for unobservable determinants of trade.

Estimation of gravity: Bilateral accessibility

Bilateral accessibility term ϕ_{ni} can include a collection of variables such as:

- distance,
- tariffs,
- free trade agreement/regional trade agreement (RTA),
- common border, language and history.

We can define a function form for ϕ_{ni} :

$$\phi_{ni} = D_{ni}^{-\theta} \eta_{ni}. \quad (18)$$

The gravity equation becomes (after taking logs):

$$\ln X_{ni} = \ln G + \ln S_i + \ln M_n - \theta \ln D_{ni} + \epsilon_{ni}. \quad (19)$$

where D_{ni} denotes the bilateral distance and $\epsilon_{ni} = \ln \eta_{ni}$. θ is the distance elasticity, which we can estimate using bilateral trade data.

Impediment to trade: Distance

The distance elasticity is stable over time and across regions.

- Using data on trade flows between countries from 1950 to 2000, the distance elasticities are estimated to be between 0.5 and 1.5 (Allen and Arkolakis, 2018).
- The distance elasticity is 1 for trade between states in the U.S (Allen and Arkolakis, 2018).
- Trade of terra sigillata during the Roman Empire exhibits a distance elasticity of 1.5 (Flückiger, Hornung, Larch, Ludwig, and Mees, 2021).
- Meta-analysis of Head and Mayer (2014): the average distance elasticity of trade is 0.93 (standard deviation: 0.4) in 159 papers (more than 2500 estimates).

Bilateral accessibility: Distance

Why can distance be used to measure (negative) bilateral accessibility/impediment to trade?

Distance is a proxy for

- transportation costs,
- iceberg trade costs,
- communication costs,
 - According to Krugman, distance “proxies for the possibilities of personal contact between managers, customers, and so on; that much business depends on the ability to exchange more information, of a less formal kind, than can be sent over a wire.”
 - Söderlund (2023) uses the liberalisation of the Soviet airspace to identify the effect of reduction in business travel time on trade.
- transaction costs.

Bilateral accessibility: Policy evaluation

- Gravity equations can be used to evaluate the effect of Regional Trade Agreement (RTA) or Free Trade Agreement.
 - We can include RTA indicators in the bilateral accessibility term.
For example:

$$\phi_{ni} = \exp(-\theta \ln D_{ni} + \gamma \text{RTA}_{ni}) \eta_{ni}. \quad (20)$$

We have:

$$\ln X_{ni} = \ln G + \ln S_i + \ln M_n - \theta \ln D_{ni} + \gamma \text{RTA}_{ni} + \epsilon_{ni}. \quad (21)$$

- As expected, trade agreements increase bilateral accessibility and substantially increase trade.
 - North American Free Trade Agreement (NAFTA) increases bilateral trade by about 50% on average.
 - European Union is associated with a 15% increase in bilateral trade on average.

Caveat: there could be endogeneity of policy implementations.

Border effects

- The Canadian-U.S. border is one of the most open borders in the world.
- The two countries are part of a free trade agreement (e.g., NAFTA, now USMCA).
 - Most goods shipped across the national border pay no tariffs.
- They share a common language.
- British Columbia has 9.5 times more trade with Ontario (in Canada) than Ohio (in the U.S.).

Is this because of distance?

Trade with British Columbia, as Percent of GDP, 2009			
Canadian Province	Trade as Percent of GDP	Trade as Percent of GDP	U.S. State at Similar Distance
Alberta	6.9	2.6	Washington
Saskatchewan	2.4	1	Montana
Manitoba	2	0.3	California
Ontario	1.9	0.2	Ohio
Quebec	1.4	0.1	New York
New Brunswick	2.3	0.2	Maine

Source: Statistics Canada, U.S. Department of Commerce.

Border effects

- High border effects exist for European trade as well.
- Why do borders make a difference after controlling for distance?
 - national institutions vary across borders (e.g., legal, monetary, social, cultural).
 - history and established business relations.

Bilateral accessibility: History and culture

- Common language:
Two countries that share a common language trade twice as much as country pairs that do not speak the same language.
- Colonial links:
Countries with colonial links trade 2.5 times as much as country pairs without colonial links.

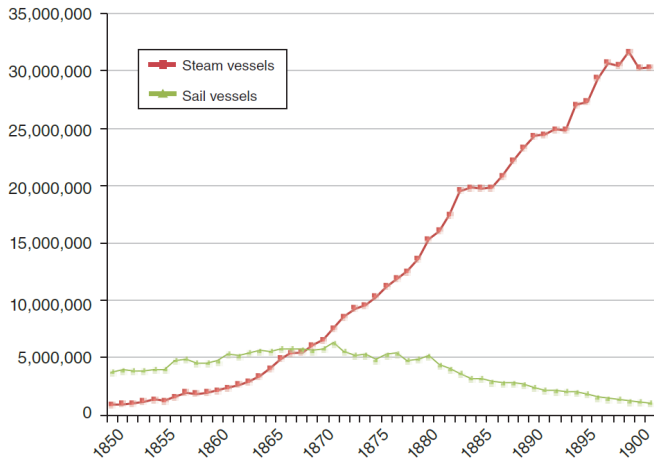
Empirical evidence

1. **The wind of change: Maritime technology, trade, and economic development** by Pascali (2017)
2. **Railroads and American economic growth: a “Market access” approach** by Donaldson and Hornbeck (2016)
3. **Shaky roads and trembling exports** by Martincus and Blyde (2013)

“The wind of change”

- Transportation costs can change due to technology change or infrastructure investment.
- Pascali (2017) looks at how the introduction of steamships in the 19th century changed world patterns of trade and shaped economic development.
- Steamships were a technological advancement compared to the incumbent: sailing ships.
 - Sailing depends heavily on wind.
 - Steamships experienced major technological breakthroughs in 1850s and 1860s.

“The wind of change”

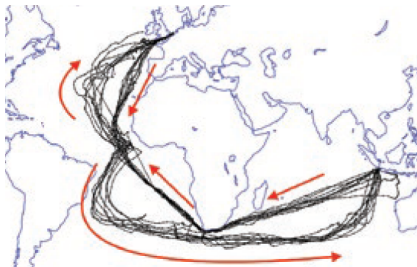


TOTAL TONNAGE OF BRITISH VESSELS ENTERED IN BRITISH PORTS FROM AND TO FOREIGN COUNTRIES AND
BRITISH POSSESSIONS

Source: Pascali (2017)

“The wind of change:” Asymmetric effect

- The advance in sea transport technology induced an asymmetric change in shipping time across trade routes.
 - e.g., The duration of a round trip for a sailing vessel from Lisbon to Cape Verde would be similar to that of around trip from Lisbon to Salvador (clockwise wind pattern in Northern Atlantic Ocean).
 - With the steamship, the duration of the former trip would be only half of the latter.



a: Shipping routes by sail



b: Optimal routes by sail

Source: Pascali (2017)

“The wind of change:” Predicted routes

The author calculates the optimal route and time for each country pair separately for two sea transport technologies (i.e. sail versus steam vessels).

- The calculation is based on Dijkstra's algorithm:
 - divide the space into cells.
 - assign cost/time to traverse each cell.
 - for sailing, the time depends on velocity/direction of wind.

“The wind of change:” Reduction in shipping time

TABLE 1—DESCRIPTIVE STATISTICS

	Mean	Median	SD	Min	Max	Observations
<i>Panel A. Unit of observation: country pair</i>						
Optimized shipping time—steam— Suez closed	434.55	397.50	250.54	0.00	1,054.48	8,834
Optimized shipping time—steam— Suez open	391.85	370.59	214.43	0.00	941.17	8,834
Optimized shipping time—sail— Suez open	914.81	857.18	513.98	0.00	2,418.92	8,834
Optimized shipping time—sail— Suez closed	855.28	818.78	461.76	0.00	2,166.01	8,834
Great-circle distance (km)	8,608.95	8,395.63	4,785.40	0.00	20,007.44	8,834

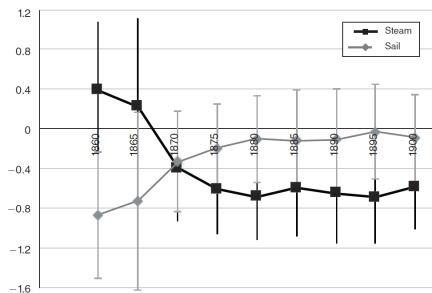
- Steamships are predicted to reduce the shipping time by more than a half on average.
- The opening of the Suez Canal is predicted to reduce the shipping time by an additional 10 percent.

“The wind of change:” from sail to steam

- The paper uses a gravity equation to determine the timing of the transition from sail to steam:

$$\ln(\text{trade}_{ijt}) = \beta_{\text{steam},T} \ln(\text{steamTIME}_{ij}) + \beta_{\text{sail},T} \ln(\text{sailTIME}_{ij}) + X_{ijt}\Gamma + \gamma_t + \epsilon_{ijt}, \quad (22)$$

where $X_{ijt}\Gamma$ include importer and exporter fixed effects (errors are three-way clustered).



The change in the elasticity of trade with respect to shipping times by sail and steam

“The wind of change:” Freight rates

Can freight rates be used instead as a proxy for changes in transportation costs?

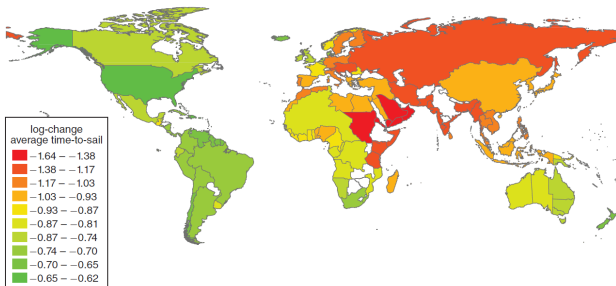
- Freight rates can respond to demand for shipping services and are also subject to the market structure of the shipping industry.
- Therefore, freight rates can be endogenous to economic development and trade activities.
- In contrast, the predicted changes in shipping times were a product of a transport technological change, and plausibly only depend on the technological parameters of the two sea transport methods.

“The wind of change:” Country-level evidence

- The paper proceeds to examine the effect of the introduction of the steamship on trade, using country-level data.
- It calculates, at the country level, the reduction in shipping time, by taking the average change in shipping times for all trading partners of a country:

$$\Delta \ln Dist_i \equiv \sum_{j \neq i} w_j [\ln(steamTIME_{ij}) - \ln(sailTIME_{ij})], \quad (23)$$

where weight w_j is country j 's share in world trade.



“The wind of change:” Country-level evidence

Then examine the following relation between the change in shipping time $\Delta \ln Dist_i$ and the change in export $\Delta \ln T_i$ between 1850 and 1905:

$$\Delta \ln T_i = c + \alpha \Delta \ln Dist_i + v_i. \quad (24)$$

- $\Delta \ln Dist_i$ captures arguably an exogenous change in shipping time, which is a function of technological changes (e.g., wind patterns).
- Therefore, the coefficient α plausibly captures the causal effect of the introduction of the steamship on trade.
- $-\alpha$ is also the distance elasticity (θ in Equation 19).

Caveat: it assumes that all international trade was carried by sail in 1850 and by steam in 1905 and this potentially generates a downward bias.

“The wind of change:” Effect on trade

	log change export/GDP			log change export/population		
	(1)	(2)	(3)	(4)	(5)	(6)
log-change distance (to all countries)	-1.008 (0.663)			-1.737 (0.480)		
log-change distance (to United Kingdom)		-0.792 (0.469)			-1.422 (0.341)	
log-change distance (to top five trading countries)			-1.498 (0.874)			-2.469 (0.662)
Intercept	-0.177 (0.582)	-0.00547 (0.436)	-0.0416 (0.441)	-0.389 (0.419)	-0.141 (0.308)	-0.100 (0.331)
R^2	0.0912	0.115	0.113	0.214	0.270	0.225
Observations	25	24	25	50	49	50
Weighted (by log population)	Yes	Yes	Yes	Yes	Yes	Yes

Table: Change in shipping time and trade

- The reduction in shipping time is associated with an increase in export/GDP ratio and export per capita.
- The elasticities are estimated to fall between 0.8 and 2.5.
- This implies that the introduction of the steamship can account for about 50% ($-0.5 \times -1.422/1.4$) of the trade boom that occurred over this period (i.e., the first wave of trade globalisation).

“Railroads and American economic growth”

- Transportation costs can also change because of infrastructure investment.
- While steamships facilitated cross-country integration, railroads played a significant role in promoting within-country integration in the 19th-century United States.
- Why is it challenging to evaluate the effects of transportation infrastructure?
 - Connection not only has a local impact, but can affect other regions indirectly through interlinked trade networks.
 - Construction of transportation infrastructure can be endogenous.
- Donaldson and Hornbeck (2016) develop the “market access” approach to study the *aggregate* effect of the expansion of railroad network.
 - they develop a framework to characterise the direct and indirect effect of railroads on each area’s market access, which can be computed/estimated using data.

“Railroads and American economic growth:” Social saving estimates

Fogel (1964) quantifies the effect of railroad expansion in the 19th century U.S. using a “social saving” approach:

- Inter-regional trade: multiply the difference in freight costs (with and without railroads) by transported agricultural goods (fixed in 1890).
- Intra-regional trade: bounded by annual rent of infeasible regions in addition.

Insights:

- Land is immobile and therefore the gains from railroads will be capitalised by land values.

Challenges:

- data limitations,
- theoretical concerns:
 - Trade in reality will respond to changes in transportation costs.

“Railroads and American economic growth:” Market access approach

- Donaldson and Hornbeck (2016) build a general equilibrium trade model with multiple regions (i.e., counties).
- A county is affected directly by being connected to the railroad, but also indirectly if other counties with which it trades are connected to the railroad → summarised in this county's “market access.”
- The model will deliver
 - a gravity equation that relates trade with market access,
 - testable predictions that are in line with Fogel's intuitions:
 - * Increase in market access → increase in agricultural land values.

“Railroads and American economic growth:” Transportation network

- The first step: use the Geographical Information System (GIS) to construct the transportation network and its change.



Waterways, canals and 1870 railways Waterways, canals and 1890 railways

- Apply Dijkstra's algorithm to calculate the optimal route and the associated transport cost between any two counties (i.e. a matrix of county-to-county transport costs)
 - by allowing transport costs to vary with waterway, wagon roads and railroad network.

“Railroads and American economic growth:” Iceberg trade cost

The bilateral transport costs are modelled in the iceberg form.

- $\tau_{od} - 1$ is the *ad valorem* transport cost.
- It is approximated by expressing the freight rates in proportional terms based on the average value of transported agricultural goods.
- The elasticity of iceberg trade costs is different from the distance elasticity in general.

“Railroads and American economic growth:” Trade model

- The second step: build a trade model that summarises the direct and indirect impacts of changing transport costs in a county's market access.

Model set-up:

- Multiple counties, a continuum of differentiated goods and CES demand.
- For trade flows, the paper denotes origin county o and destination county d .
- Perfect competition and Cobb-Douglas technology to produce varieties from labour, capital and land.
- Capital and labour are perfectly mobile. Land is immobile.

We will see how to solve this class of models in Lecture 4. Here we instead focus on the predictions of the model and how to use market access in the empirical estimation.

“Railroads and American economic growth:” Market access

Equation 6 in Donaldson and Hornbeck (2016)

$$(1 + \alpha\theta) \ln q_o = \kappa_2 + \ln\left(\frac{A_o}{L_o}\right) - \gamma\theta \ln \bar{U} + \gamma \ln CMA_o + \ln FMA_o, \quad (25)$$

where q_o is the land rental rate; CMA_o and FMA_o are consumer and firm market access, respectively.

- FMA_o is the sum of the size of the destination market (Y_d), inversely weighted by its competitiveness (CMA_d) and the bilateral cost of trading (τ_{od}):

$$FMA_o = \sum_d \tau_{od}^{-\theta} CMA_d^{-1} Y_d. \quad (26)$$

- Similarly, CMA_d is the weighted sum of production from different sources:

$$CMA_d = \sum_o \tau_{od}^{-\theta} FMA_o^{-1} Y_o. \quad (27)$$

Note: compare to Equation 6.

“Railroads and American economic growth:” Market access

Under the assumption that trade costs are symmetric (i.e., $\tau_{od} = \tau_{do}$), FMA and CMA are equal to one another up to a proportionality ($MA = FMA = \rho CMA$). We can further simply and write:

$$MA_o = \kappa_3 \sum_d \tau_{od}^{-\theta} MA_d^{\frac{-(1+\theta)}{\theta}} N_d, \quad (28)$$

where N_d is the number of workers. A county's market access (MA) is the weighted sum of the populations of the counties it trades with, giving a lower weight for counties:

- with whom it is costly to trade with (high τ_{od}),
- that have itself a large MA .

“Railroads and American economic growth:” Market access

- Market access measures firms' desire to sell goods elsewhere for a high price and consumers' desire to buy goods from elsewhere at a low price.
- Both direct and indirect effects of railroads will be captured by changes in market access. New railroad connection to county A affects:
 - county A,
 - those can now (more easily) trade with A,
 - those that had been trading with A,
 - those that had traded with A's previous trade partners,
 - those that had traded with county A's new trade partners,
 - and so on.
- If we instead compare counties with new railroad connection and counties without, the estimated effect will be biased because of the general equilibrium effect through the trade network.

“Railroads and American economic growth:” Prediction

Given MA , Equation 25 becomes:

$$\ln q_o = \kappa_4 + \frac{1}{1 + \alpha\theta} \ln \frac{A_o}{L_o} + \frac{1 + \gamma}{1 + \alpha\theta} \ln MA_o. \quad (29)$$

Equilibrium rental rates are log-linear in market access. Immobile land in county o will be more valuable if it has cheaper access to

- large uncompetitive markets,
- labour.

“Railroads and American economic growth:” Estimation

Challenge 1:

- County productivity (A_o) is not observed in data.
- The paper assumes that changes in county productivity are orthogonal to changes in market access from 1870 to 1890, after controlling for counties geographic location (state, longitude, latitude).
 - Why do we need to make this assumption?
 - Why control state, longitude and latitude?

“Railroads and American economic growth:” Estimation

Challenges 2&3:

- Market access (MA_o) is not directly observed in data.
- We can solve numerically MA_o using Equation 28.
- Alternatively, the paper uses the following first-order approximation:

$$MA_o \approx \sum_d \tau_{od}^{-\theta} N_d. \quad (30)$$

where trade elasticity θ is estimated to be 8.22.

- The population N_o is endogenously co-determined with the land rental rate q_o .
 - Each county's own population is excluded from its measure of market access.
 - For robustness checks, nearby counties are also excluded to mitigate the effect of local shocks.

“Railroads and American economic growth:” Estimation

The estimation equation:

$$\ln V_{ot} = \beta \ln MA_{ot} + \delta_o + \delta_{st} + f(x_o, y_o)\delta_t + \epsilon_{ot}, \quad (31)$$

where $f(x_o, y_o)$ is a cubic polynomial in county latitude and longitude.

- Baseline weight by counties' land value in 1870.
- Standard errors are clustered at the state level.

“Railroads and American economic growth:” Results

	Log Value of Agricultural Land				
	(1)	(2)	(3)	(4)	(5)
	Baseline	Model-Derived	Fixed	100-Mile	
	Specification	Market	1870	Buffer	
		Access	Population	Market	Unweighted
				Access	
Log market access	0.511	0.587	0.510	0.487	0.506
	(0.065)	(0.073)	(0.065)	(0.064)	(0.124)
Number of counties	2,327	2,327	2,327	2,327	2,327
<i>R</i> -squared	0.625	0.627	0.625	0.621	0.606

Table: Impact of market access on agricultural land value

“Railroads and American economic growth:” Endogeneity

- Railroad expansion may not be random, which can cause an endogeneity problem:
 - e.g., railroad construction could favour counties where agricultural land values were trending upward.
- The use of market access helps because
 - it captures both the direct and indirect effects of railroad expansion.
- But it cannot completely eliminate the endogeneity problem.
 - Solution 1 of the paper: Control local railroads.
 - Solution 2 of the paper: Instrumental variable (IV) estimation using “water market access.”
 - * The gains from railroad construction will be smaller for counties already with good access to navigable waterways.
 - * What is the other underlying assumption for the IV to be valid?

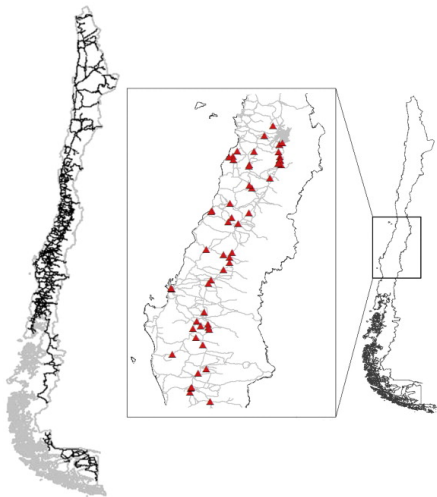
“Railroads and American economic growth:” Summary

- It examines the *aggregate* effect of railroads on the U.S. economy with
 - a general equilibrium trade model that derives a gravity equation relating land values to market access,
 - county-to-county transport cost matrix to compute each county's market access.
- Under the counterfactual that railroads were not extended, the agricultural land values would have dropped by 60%.
- The effect is robust to controlling for regional changes in agricultural land and alternative population distributions.
- The effect of alternative transportation infrastructure would have been limited.
- It mentions the possibility that the U.S. railroads can even affect prices, wages and worker utility in the rest of the world via international trade and migration.

“Shaky roads and trembling exports”

- Martincus and Blyde (2013) look at a negative shock to the domestic transportation infrastructure in Chile and examine how this affected its foreign export.
- On February 27, 2010 an earthquake with a magnitude of 8.8 on the moment magnitude scale took place in Chile.
- The earthquake caused severe damage to the transportation network (9% paved road network).
- Repair took several months.

“Shaky roads and trembling exports”

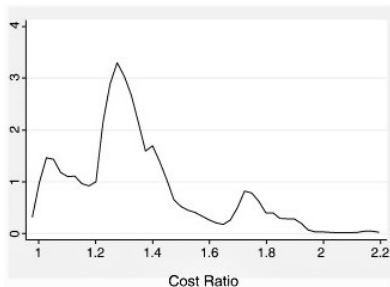


Chile's primary and secondary road network and the traffic interruption points (red triangles).

Source: Martincus and Blyde (2013)

“Shaky roads and trembling exports:” Identification

- Firms presumably use least-cost-routes to transport goods from the production facility to the existing customs for foreign export.
- Some of the routes became impassable due to the destruction caused by the earthquake.
- As a result, some firms, but not others, had to reroute to alternative routes, causing an increase in the transportation costs.



Distribution of (new cost/old cost)

Source: Martincus and Blyde (2013)

“Shaky roads and trembling exports:” Identification

- This provides plausibly exogenous variations in transport costs to identify its effect on export trade.
 - What can be potential threats to the identification?
- The baseline estimation follows the specification:

$$\Delta \ln X_{fpc} = \alpha D_{fpc} + \delta'_f + \rho'_{pc} + \epsilon_{fpc} \quad (32)$$

where X_{fpc} denotes export of product p from firm f to country c and D_{fcp} is an indicator for rerouting (predicted).

- This is essentially a difference-in-difference estimation that compares the change in export for those whose routes were damaged and whose were not.
- Fixed effects account for systematic change across firms and product-destination shocks before and after the earthquake.

“Shaky roads and trembling exports:” Results

Impact of diminished domestic road infrastructure on firms' exports

Sample: manufacturing firms, February 27, 2009-February 26, 2011

	D > 0	D > 50	D = 100
D	-0.411	-0.474	-0.417
Heteroscedasticity-consistent	(0.071)***	(0.131)***	(0.143)***
Cluster-region	(0.138)***	(0.221)**	(0.227)*
Cluster- <i>comuna</i>	(0.147)***	(0.198)**	(0.201)**
Cluster-sector ISIC 2 digit	(0.089)***	(0.081)***	(0.088)***
Cluster-sector ISIC 4 digit	(0.097)***	(0.208)**	(0.185)**
Cluster-region-sector ISIC 2 digit	(0.115)***	(0.198)**	(0.202)**
Cluster-region-sector ISIC 4 digit	(0.122)***	(0.199)**	(0.202)**
Cluster- <i>comuna</i> -sector ISIC 2 digit	(0.147)***	(0.195)**	(0.197)**
Cluster- <i>comuna</i> -sector ISIC 4 digit	(0.147)***	(0.193)**	(0.195)**
Firm fixed effect	Yes	Yes	Yes
Product-destination fixed effect	Yes	Yes	Yes
Observations	20,171	19,042	18,823

- Export that had to be rerouted suffers a 33.7% ($\exp^{-0.411} - 1$) lower increase.

“Shaky roads and trembling exports:” Threats to identification

1. The earthquake can affect local production facilities.
 - This is not a problem if it affected firm exports for all destinations similarly (with a firm fixed effect).
 - It will be problematic if it disproportionately affected firm exports with impassable routes.
 - How to solve the problem?
2. In the absence of the earthquake, affected and non-affected exports may not follow parallel trends.
 - How to test this?
3. Rerouting of shipments may cause congestion even for firms whose export routes were not affected.
 - If this were the case, how would the estimate be affected?

“Shaky roads and trembling exports:” Heterogeneity

- The previous analysis estimated an average effect for all firms.
- However, the effect can be heterogeneous for different types of firms and products. This can be estimated with the following equation:

$$\Delta \ln X_{fpc} = \sum_{i=1}^I \alpha_i \Theta_i D_{fpc} + \delta'_f + \rho'_{pc} + \epsilon_{fpc} \quad (33)$$

where Θ_i is a binary indicator for group i .

- The paper finds that the negative effect of the earthquake is particularly strong for larger firms exporting non-differentiated goods.

Conclusions

- Gravity equations can be used to characterise patterns of trade and evaluate the effect of trade policies.
- According to gravity equations, a reduction of trade barriers will lead to an increase in bilateral trade.
 - Trade costs can change when transport technologies change (e.g., introduction of the steamship) or transport infrastructure change (e.g., earthquake in Chile).
- Gravity equations can be micro-founded by a large set of trade models and they highlight the general equilibrium effect of transport infrastructure investment.
 - When a location gains access to trade with other locations, its trading partners is also affected indirectly, along with the partners of those trading partners, and so on.
 - The aggregate effect can be summarised in “market access.”

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